

Modeling, Proposed Control and Adaptive RBF Network of a 2 MW Direct-Drive PMSG Wind Energy Conversion System

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Sections II–IV of the paper. Receptor (motor) convention for the PMSG; all errors are defined as measured minus reference values. The simulation code is `matlab_v2/PMSG_WECS.m`.

2 Wind Turbine Model

The power captured by the rotor and its tip-speed ratio are [16]

$$P_w = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3, \quad \lambda = \frac{\Omega R}{v} \quad (1)$$

where ρ is the air density, R the rotor radius, v the wind speed, Ω the rotor speed and β the pitch angle ($\beta = 0$ below rated wind). The power coefficient is

$$C_p = 0.5176 \left(\frac{116}{\lambda_i} - 0.4\beta - 5 \right) e^{-21/\lambda_i} + 0.0068\lambda, \quad (2)$$

with $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3 + 1)$; its maximum $C_{p,\max} = 0.48$ is reached at $\lambda_{opt} = 8.1$. The aerodynamic torque is $T_w = P_w/\Omega$ and the direct-drive shaft obeys

$$J \frac{d\Omega}{dt} = T_w + T_e - B\Omega \quad (3)$$

where J is the total inertia, B the friction coefficient and T_e the electromagnetic torque in the receptor convention ($T_e < 0$ in generating mode).

The wind speed combines a mean value, the low-frequency harmonics of [14] and a turbulence:

$$v(t) = V_m + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_t(t) \quad (4)$$

with $V_m = 8.5$ m/s, $A = [1.5, 0.2, 0.3]$ m/s and $f = [0.03, 0.16, 0.17]$ Hz. The rotor-effective turbulence v_t is a first-order coloured noise with standard deviation 0.5 m/s (6 % of V_m) and correlation time 1.5 s. The wind used by the controller is measured by an anemometer modelled by a first-order lag of 1 s, v_m .

3 PMSG, Converters and Grid

3.1 PMSG

In the rotor dq frame (amplitude-invariant transformation, d axis on the magnet flux), with the receptor convention and $L_d = L_q = L_s$ [4],

$$v_{sd} = R_s i_{sd} + L_s \frac{di_{sd}}{dt} - \omega_e L_s i_{sq} \quad (5)$$

$$v_{sq} = R_s i_{sq} + L_s \frac{di_{sq}}{dt} + \omega_e L_s i_{sd} + \omega_e \psi_f \quad (6)$$

$$T_e = \frac{3}{2} p \psi_f i_{sq} \quad (7)$$

where $\omega_e = p\Omega$, p is the number of pole pairs and ψ_f the magnet flux. In generating mode $i_{sq} < 0$. With $\mathbf{i}_s = [i_{sd} \ i_{sq}]^T$, $\mathbf{v}_s = [v_{sd} \ v_{sq}]^T$, $\mathbf{e}_q = [0 \ 1]^T$ and the skew-symmetric matrix $\mathcal{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$,

$$L_s \dot{\mathbf{i}}_s = \mathbf{v}_s - R_s \mathbf{i}_s + \omega_e L_s \mathcal{J} \mathbf{i}_s - \omega_e \psi_f \mathbf{e}_q. \quad (8)$$

3.2 Back-to-Back Converter, DC Link and Grid

The two-level converters are represented by their average model; the voltage vectors are limited to $V_{dc}/\sqrt{3}$ (space-vector modulation). The DC link receives the power of the machine-side converter (MSC) and supplies the grid-side converter (GSC):

$$CV_{dc}\frac{dV_{dc}}{dt} = P_{dc} - P_i, \quad P_{dc} = -\frac{3}{2}\mathbf{v}_s^T \mathbf{i}_s, \quad P_i = \frac{3}{2}\mathbf{v}_i^T \mathbf{i}_g. \quad (9)$$

In the grid-voltage-oriented frame ($v_{gq} = 0$, ideal PLL), the L filter between the GSC voltage $\mathbf{v}_i$ and the grid voltage $\mathbf{v}_g$, with the current $\mathbf{i}_g$ counted towards the grid, gives

$$L_f \dot{\mathbf{i}}_g = \mathbf{v}_i - R_f \mathbf{i}_g + \omega_g L_f \mathcal{J} \mathbf{i}_g - \mathbf{v}_g, \quad (10)$$

and the grid powers are $P_g = \frac{3}{2}v_{gd}i_{gd}$ and $Q_g = -\frac{3}{2}v_{gd}i_{gq}$.

3.3 Power Balance

Equations (3), (8)–(10) form a sixth-order model with state $[i_{sd}, i_{sq}, \Omega, V_{dc}, i_{gd}, i_{gq}]$. With the stored energy $E = \frac{1}{2}J\Omega^2 + \frac{3}{4}L_s\|\mathbf{i}_s\|^2 + \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2$, and since $\mathbf{x}^T \mathcal{J} \mathbf{x} = 0$ and $T_e \Omega = \frac{3}{2}\omega_e \psi_f i_{sq}$, the model gives exactly

$$\dot{E} = P_w - B\Omega^2 - \frac{3}{2}R_s\|\mathbf{i}_s\|^2 - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 - P_g \quad (11)$$

for any state and converter voltages: the aerodynamic power is stored, dissipated or delivered to the grid. This identity checks every sign of the model; numerically, its residual is below $10^{-15}P_n$.

Table 1 lists the parameters. The turbine data (R, Ω_n, T_n) correspond to a 2 MW direct-drive turbine; the generator, DC-link, filter and grid data are those of the non-salient PMSG of [15]. The inertia gives an inertia constant $H = J\Omega_n^2/(2P_n) = 3.5$ s. At the rated point ($i_{sq} = -2.51$ kA, $\omega_e = 58$ rad/s), the MSC voltage is 577 V and the GSC voltage 579 V, below the limit $V_{dc}/\sqrt{3} = 693$ V.

Table 1: Parameters of the Studied System

Parameter	Value	Parameter	Value
P_n	2 MW	p	26
R	40 m	R_s	0.78 mΩ
Ω_n, v_n	2.23 rad/s, 11.0 m/s	L_s	1.57 mH
T_n	0.897 MN·m	ψ_f	9.18 Wb
$\lambda_{opt}, C_{p,max}$	8.1, 0.48	C, V_{dc}^*	20 mF, 1200 V
J	2.8×10^6 kg·m ²	L_f, R_f	0.15 mH, 2 mΩ
B	10^3 N·m/s/rad	Grid	690 V, 50 Hz
ρ	1.225 kg/m ³	f_{sw}	2160 Hz

4 Proposed Control

The structure is shown in Fig. 1. All errors are defined as measured minus reference values. The controller is sampled at $T_s = 1/(2f_{sw}) = 231.5 \mu\text{s}$ (regular sampling, double update).

4.1 Speed Reference

The optimal speed $\Omega_{opt} = \lambda_{opt}v_m/R$, limited to $[0.5\Omega_n, \Omega_n]$, is filtered by $\ddot{\Omega}^* = \omega_r^2(\Omega_{opt} - \Omega^*) - 2\omega_r\dot{\Omega}^*$ ($\omega_r = 1$ rad/s), which provides a smooth reference Ω^* and its derivative.

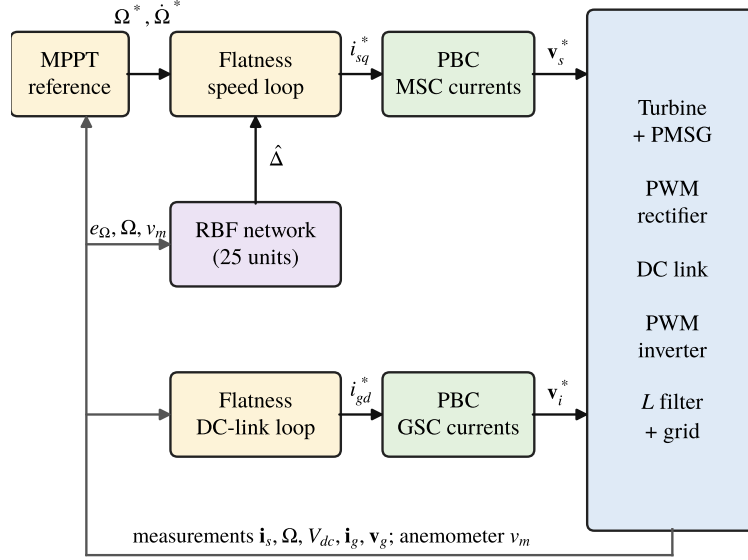


Figure 1: Control structure: flat outer loops, PBC current loops and RBF compensation.

4.2 Flatness-Based Speed Loop

A system is flat if its states and inputs are functions of a flat output and of a finite number of its derivatives [10]. Flatness is used here for the two outer subsystems, with the currents as inputs. From (3), $T_e = J\dot{\Omega} + B\Omega - T_w$: Ω is a flat output of the shaft with input T_e , i.e., i_{sq} . Imposing $\dot{\Omega} = v_1$ with $v_1 = \dot{\Omega}^* - k_{11}e_\Omega - k_{12}\int e_\Omega dt$, $e_\Omega = \Omega - \Omega^*$, gives

$$T_e^* = Jv_1 + B\Omega - \hat{T}_w - \hat{\Delta}, \quad i_{sq}^* = \frac{T_e^*}{\frac{3}{2}p\psi_f}, \quad i_{sd}^* = 0 \quad (12)$$

where $\hat{T}_w$ is the aerodynamic torque computed from (1)–(2) with v_m and $\hat{\Delta}$ is the neural compensation (Section IV-E). T_e^* is limited to $[-1.2T_n, 0]$ and the integral is frozen at the limit. With $i_{sq} = i_{sq}^*$ and $\Delta = T_w - \hat{T}_w$,

$$J(\dot{e}_\Omega + k_{11}e_\Omega + k_{12}\int e_\Omega dt) = \Delta - \hat{\Delta}. \quad (13)$$

4.3 Flatness-Based DC-Link Loop

The flat output is the energy stored in the DC link and in the filter, $y_2 = \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2$. From (9)–(10),

$$\dot{y}_2 = P_{dc} - P_g - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 \quad (14)$$

which is linear in the grid power [11]; V_{dc} follows from y_2 and the measured currents, and P_g from $\dot{y}_2$. Imposing $\dot{y}_2 = -k_{21}e_y - k_{22}\int e_y dt$, with $e_y = y_2 - y_2^*$ and $y_2^* = \frac{1}{2}CV_{dc}^{*2} + \frac{3}{4}L_f\|\mathbf{i}_g^*\|^2$, gives

$$P_g^* = P_{dc} - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 + k_{21}e_y + k_{22}\int e_y dt, \quad i_{gd}^* = \frac{2P_g^*}{3v_{gd}} \quad (15)$$

and $i_{sq}^* = -2Q_g^*/(3v_{gd})$ with $Q_g^* = 0$. P_{dc} is computed from the applied MSC voltages and the measured currents, so the power coming from the generator is compensated before it changes V_{dc} . The variation of y_2^* , due to the filter energy only (below 300 J against 14.4 kJ in the capacitor), is neglected. The gains are $k_{11} = 2\omega_\Omega$, $k_{12} = \omega_\Omega^2$, $k_{21} = 2\omega_v$, $k_{22} = \omega_v^2$ (Table 2).

4.4 Passivity-Based Current Loops

With $\mathbf{e}_s = \mathbf{i}_s - \mathbf{i}_s^*$ and $\dot{\mathbf{z}}_s = \mathbf{e}_s$, the MSC voltage reference is

$$\mathbf{v}_s^* = L_s\dot{\mathbf{i}}_s^* + R_s\mathbf{i}_s^* - \omega_e L_s \mathcal{J} \mathbf{i}_s^* + \omega_e \psi_f \mathbf{e}_q - R_a \mathbf{e}_s - K_i \mathbf{z}_s \quad (16)$$

and, with $\mathbf{e}_g = \mathbf{i}_g - \mathbf{i}_g^*$ and $\mathbf{z}_g = \mathbf{e}_g$, the GSC voltage reference is

$$\mathbf{v}_i^* = L_f \dot{\mathbf{i}}_g^* + R_f \mathbf{i}_g^* - \omega_g L_f \mathcal{J} \mathbf{i}_g^* + \mathbf{v}_g - R_b \mathbf{e}_g - K_{ig} \mathbf{z}_g. \quad (17)$$

The reference derivatives are obtained by first-order filters (2 ms); the integrals are frozen when the voltage reaches $V_{dc}/\sqrt{3}$.

Proposition 1. *With (16) applied to (8) and $R_a, K_i > 0$, the storage function $H = \frac{1}{2} L_s \|\mathbf{e}_s\|^2 + \frac{1}{2} K_i \|\mathbf{z}_s\|^2$ satisfies $\dot{H} = -(R_s + R_a) \|\mathbf{e}_s\|^2$ and $\mathbf{e}_s \rightarrow 0$. The same holds for (17) applied to (10).*

Proof. Subtracting $L_s \dot{\mathbf{i}}_s^*$ from (8) and inserting (16) gives $L_s \dot{\mathbf{e}}_s = -(R_s + R_a) \mathbf{e}_s + \omega_e L_s \mathcal{J} \mathbf{e}_s - K_i \mathbf{z}_s$. Hence $\dot{H} = \mathbf{e}_s^T L_s \dot{\mathbf{e}}_s + K_i \mathbf{z}_s^T \dot{\mathbf{e}}_s = -(R_s + R_a) \|\mathbf{e}_s\|^2$, since $\mathbf{e}_s^T \mathcal{J} \mathbf{e}_s = 0$: the coupling is workless. H is bounded and $\dot{H}$ is uniformly continuous, so Barbalat's lemma [17] gives $\mathbf{e}_s \rightarrow 0$. $\square$

The gains place a double closed-loop pole at ω_i : $R_s + R_a = 2\omega_i L_s$, $K_i = \omega_i^2 L_s$, $R_f + R_b = 2\omega_i L_f$, $K_{ig} = \omega_i^2 L_f$. With $\omega_i = 500$ rad/s, the current loops are 250 times faster than the speed loop and 8 times faster than the DC-link loop.

4.5 Adaptive RBF Compensation

The torque model error $\Delta = T_w - \hat{T}_w$ comes from the C_p model, the anemometer and the generator parameters. On the operating region it is approximated by $\Delta = \mathbf{W}^* \phi(\mathbf{z}) + \varepsilon$, $|\varepsilon| \leq \bar{\varepsilon}$, where $\mathbf{z} = [\Omega/\Omega_n, v_m/v_n]^T$ and ϕ contains 25 Gaussian functions $\phi_j = \exp(-\|\mathbf{z} - \mathbf{c}_j\|^2/(2b^2))$ with centres on a 5×5 grid of $[0.5, 1.1]^2$ and $b = 0.15$. The compensation $\hat{\Delta} = \hat{\mathbf{W}}^T \phi$ replaces the integral of the speed loop ($k_{12} = 0$ when the network is used) and is adapted by

$$\dot{\hat{\mathbf{W}}} = \Gamma(\phi e_\Omega - \sigma \hat{\mathbf{W}}), \quad (18)$$

frozen when T_e^* saturates, with $\hat{\mathbf{W}}(0) = 0$.

Proposition 2. *Consider (13) with $k_{12} = 0$, the law (18) with $\Gamma, \sigma > 0$, exact J and B , and no saturation. Then e_Ω and $\tilde{\mathbf{W}} = \mathbf{W}^* - \hat{\mathbf{W}}$ are uniformly ultimately bounded; the residual set shrinks when k_{11} increases and when $\bar{\varepsilon}$ and $\sigma \|\mathbf{W}^*\|^2$ decrease.*

Proof. With $V = \frac{1}{2} J e_\Omega^2 + \frac{1}{2\Gamma} \|\tilde{\mathbf{W}}\|^2$ and (13), $\dot{V} = -J k_{11} e_\Omega^2 + e_\Omega (\tilde{\mathbf{W}}^T \phi + \varepsilon) - \frac{1}{\Gamma} \tilde{\mathbf{W}}^T \dot{\tilde{\mathbf{W}}} = -J k_{11} e_\Omega^2 + e_\Omega \varepsilon + \sigma \tilde{\mathbf{W}}^T \tilde{\mathbf{W}}$. Using $\tilde{\mathbf{W}}^T \tilde{\mathbf{W}} \leq \frac{1}{2} \|\mathbf{W}^*\|^2 - \frac{1}{2} \|\tilde{\mathbf{W}}\|^2$ and $e_\Omega \varepsilon \leq \frac{J k_{11}}{2} e_\Omega^2 + \frac{\bar{\varepsilon}^2}{2J k_{11}}$ gives $\dot{V} \leq -\frac{J k_{11}}{2} e_\Omega^2 - \frac{\sigma}{2} \|\tilde{\mathbf{W}}\|^2 + \frac{\bar{\varepsilon}^2}{2J k_{11}} + \frac{\sigma}{2} \|\mathbf{W}^*\|^2$, which is negative outside a compact set. $\square$

The result is local: it assumes fast current loops, no saturation and $\mathbf{z}$ in the region covered by the centres; mismatches of J and B add bounded terms to ε . The PBC loops and the DC-link loop are not modified by the network.

Table 2: Controller Parameters

Loop	Parameters
Speed (flat)	$\omega_\Omega = 2$ rad/s ($k_{11} = 4, k_{12} = 4$), $\omega_r = 1$ rad/s
DC link (flat)	$\omega_v = 60$ rad/s ($k_{21} = 120, k_{22} = 3600$)
PBC MSC	$\omega_i = 500$ rad/s: $R_a = 1.57 \Omega, K_i = 393 \Omega/\text{s}$
PBC GSC	$\omega_i = 500$ rad/s: $R_b = 0.148 \Omega, K_{ig} = 37.5 \Omega/\text{s}$
RBF network	25 units, $b = 0.15, \Gamma = 10^7, \sigma = 0.02 \text{ s}^{-1}$

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